Phase I Swindon EV

Sustainability Report

Swindon EDA Consortium

Brookbanks

This Page Is Intentionally Blank

Document Control Sheet

Document Title: Sustainability Report

Document Ref: 1335 RE 01

Project Name: Phase 1 Swindon EV

Project Number: 1335

Client: Swindon EDA Consortium

Document Status

Rev	Issue Status	Prepared / Date	Checked / Date	Approved / Date
0	Draft	CA 10.07.13	DW 11.07.13	PAB 12.07.13
1	Final	CA 10.10.13	DW 11.10.13	PAB 14.10.13
2	Final	CA 15.10.13	DW 15.10.13	PAB 15.10.13

Issue Record

Name / Date & Revision	10.07.13	14.10.13	15.10.13			
Mr N Duckworth – Hallam land Management	0	1	2			
Mr A Birch – Hallam Land Management	0	1	2			
Mr J Aylwin – Taylor Wimpey	0	1	2			
Mr D Woodward – Hanick Homes	0	1	2			

© Copyright Brookbanks Consulting Ltd 2013
This document may not be reproduced or transmitted, in any form or by any means whether electronic, mechanical, photographic, recording or otherwise, or stored in a retrieval system of any nature without the written permission of Brookbanks Consulting Limited. No part of this work may be modified without the written permission of Brookbanks Consulting Ltd. No part of this work may be exposed to public view in any form or by any means, without identifying the creator as Brookbanks Consulting Ltd.

This Page Is Intentionally Blank

Contents

1	Introduction	1
2	Background Information	1
3	Policy Review	2
4	Development Sustainability/ Energy Proposals	6
5	Code for Sustainable Homes	6
6	Sustainability Benefits	9
7	Energy Statement	11
8	Summary	16
9	Limitations	16

This Page Is Intentionally Blank

1 Introduction

1.1 Brookbanks Consulting Limited has been appointed by the Swindon EDA Consortium to complete a Sustainability Report for the proposed residential development consisting of Phase I of the Swindon New Eastern Villages.

1.2 This report considers potential sustainability measures available to the development to meet national requirements while allowing for the practical implementation of the proposals in relation to the current and emerging building standards, including compliance with Code for Sustainable Homes and the Building Regulations.

1.3 The report therefore reviews technologies and systems that will ensure the development delivers a superior sustainable settlement that encompass measures meeting UK Building Regulations and reviewing policy requirements.

1.4 Further to this, the report goes on to outline wider sustainability benefits for the site and the existing areas, providing a commentary on a number of points including transport, waste and drainage.

2 Background Information

Location & Details

2.1 The proposed development lies to the east of Swindon. The A420 and the Great Western Rail Line bound the site to the south. To the west of the site, Thornhill Lane forms the site boundary, with the extents of South Marston, Nightingale Lane and the accompanying farmland bounding the site to the north. Agricultural land forms the eastern boundary of the site.

2.2 The majority of the land is thought to be historically undeveloped, however, small developments including a combination of dwellings, hotels and public houses are spread around the site, with link roads included throughout. The site location and boundary is shown indicatively on Figure 2a, below.

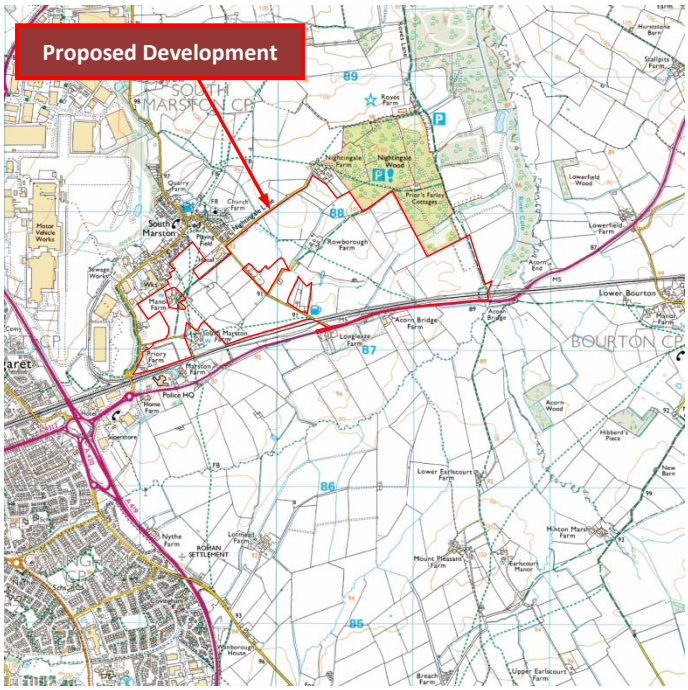

Figure 2a: Site location

Development Criteria

2.3 It is proposed to develop circa 2,380 residential dwellings within the 162 ha site.

3 Policy Review

3.1 To fully consider the likely obligations and implications of providing a sustainable development through the introduction of high quality sustainable technologies, it is necessary firstly to appraise the present and emerging policy criteria. This section of the report reviews National and Local Policy guidance, to understand the minimum site wide requirements in terms of sustainability and associated implications.

3.2 The National Planning Policy Framework (NPPF) defines the overarching aims of the Government's sustainable development strategy.

3.3 NPPF outlines that local authorities should adopt proactive strategies to mitigate and adapt to climate change and that to support the move to a low carbon future local authorities should:

- plan for new development in locations and ways which reduce greenhouse gas emissions;
- actively support energy efficiency improvements to existing buildings; and
- when setting any local requirement for a building's sustainability, do so in a way consistent with the Government's zero carbon buildings policy and adopt nationally described standards.

3.4 The NPPF stresses that the importance of sustainability in new developments is to ensure that during construction and operation the development minimises environmental impact. The Government is keen to limit the environmental impact of new construction projects through the reduction of CO_2 emissions. Subsequently it is imperative all new developments demonstrate sufficiently that no significant negative impact will be caused on what is regarded as the acceptable baseline, Building Regulation standards, and how new developments will look to achieve this.

3.5 Additional national policy goes on to specify standards and solutions in achieving further criteria associated with sustainable development, such as CO_2 and energy use reductions.

3.6 The significance of renewable energy, low carbon and energy efficiency technologies is stressed throughout national policy with the intention of introducing into local policy to accord with the national consensus.

3.7 National Policy for Low Carbon and Renewable Energy technology is informed by:

- National Planning Policy Framework (2012) Published by the UK Government.
- Code for Sustainable Homes. (2010) Published by Building Research Establishment.
- The Energy White Paper (2003 & 2007). Published by the UK Government.
- The Renewable Energy Strategy (2009). Published by the UK Government.
- Building Regulations Part L (2010). Published by the UK Government.

3.8 The main aim of these documents is to inform policy and provide guidelines to reduce the UK CO_2 emissions, as this is currently considered to be the largest man made contributor to climate change. The UK Government has set challenging targets for generating electricity from renewable sources. By 2015 15% of electricity generation must be from renewable sources and by 2020 a total of 20% (Energy White Paper : 2003)

3.9 In 2009 the UK Renewable Energy Strategy reviewed the current targets and reaffirmed the methodology for the UK achieving a 15% share of total energy from renewable energy sources by 2020. Within this document, renewable technologies for electricity, heat and transport were considered in reaching the 15% target.

3.10 As part of the UK's drive for delivering sustainable development the BRE have introduced a nationally led home assessment methodology to rate the level of sustainability within proposed new dwellings. The Code for Sustainable Homes standards have been introduced as a mandatory standard in the construction of affordable housing schemes nationwide and in many cases has been introduced to local and regional policies as a requirement in the construction of new homes. These standards promote more sustainable construction techniques in the implementation of new builds as well as limiting the environmental impact.

3.11 This document will further go on to discuss the implications of Code for Sustainable Homes and the likely obligations that are imposed upon new developments to meet these standards.

3.12 Following the amendment of the Building Regulations in October 2010 a further carbon emissions reduction of 25% from the 2006 Building Regulations Part L standard was implemented. Subsequently, to achieve the Code for Sustainable Homes Level 3, minimum standard for CO_2 emissions, and compliance with Building Regulations Part L: 2010 was required. In the future it is likely Part L of the Building Regulations will regularly be updated in line with the carbon reduction targets of Code for Sustainable Homes. For instance Building Regulations Part L: 2013 includes further requirements for increased energy efficiency, which results in a further reduction of approximately 11%.

Local Policy

3.13 Under the transitional agreements of the Planning and Compulsory Purchase Act, the Swindon Borough Local Plan (2001-2011) was adopted for a period of 3 years ending 19th July 2009. A number of policies within this local plan have been saved however, some policies were superseded by the central area action plan in February 2009.

3.14 The Swindon Borough Local Plan 2026 was submitted to the Secretary of State on 28th June 2013. The submitted Local Plan provides text on High Quality Design and Sustainable Development, with Policy DE1 as follows:

Policy DE1: High Quality Design

a. *"High standards of design are required for all types of development. Proposals for development should address the objectives of sustainable development through high quality design and place-making principles. To ensure this, proposals will be assessed against all the following design principles:*
   - *Context and character,*
   - *Layout, form and function of the development, including;*
     - *Accessibility;*
     - *Connectivity;*
     - *Legibility;*
     - *Inclusivity;*
     - *Safety and Security;*
     - *Efficiency;*
     - *Permeability; and*
     - *Adaptibility;*
   - *Amenity; and*
   - *Quality of the Public Realm"*

Figure 3a: Policy DE1 – High Quality Design

3.15 The submitted Local Plan breaks down this policy and explains the requirements in further detail below:

Character and Context

4.9 *"Each place has its own distinctive character and possesses inherent design and assets and opportunities. As such, all development must be in context with the existing natural, built and historic environment and proposals must respond positively to enhance or create distinctive character and identity."*

Layout, Form and Function of Development

4.14 *"Development layouts should aim to achieve the following principles as follows:*

- ***Accessibility*** *by all modes of transport giving priority to pedestrians and cyclists;*
- ***Integration*** *between new development and the existing context and character relating to the historic environment; urban structure; urban grain; green infrastructure; public realm; and public and community facilities;*
- ***Legibility*** *– Through a clear and logical structure that is easy to navigate and understand. This should be achieved through a well-ordered and defined public realm, with clear relationships between uses, buildings, routes and spaces;*
- ***Inclusivity*** *across all user groups through the principles of inclusive design. Inclusive access to places, spaces, buildings and public transport is a crucial part of achieving social equality and the prevention of economic, social and physical barriers;*
- ***Safety & Security*** *– Development must be safe, secure and attractive to minimise opportunities for criminal activity and reduce the fear of crime. This will be assessed in terms of mix of uses, active frontages onto streets, access routes and routes through the development (including integrated routes), public realm, surveillance, lighting, clear entrance points and the location and extent of planting, paying particular attention to the needs of those with disabilities or restricted mobility;*
- ***Efficiency*** *– in design and layout to optimise energy efficiency through passive solar design considerations; and in the use of land without compromising existing and proposed amenity;*
- ***Permeability*** *– by creating development which is well connected to the surrounding movement network and which enables ease of movement within and across the site; and*
- ***Adaptability*** *– in the robustness of the pattern, arrangement, size and configuration of plots and blocks of development and of building design to allow for change of the long term."*

Amenity

4.17 *"Development proposals must ensure convenience and levels of comfort and enjoyment, and promote good health and well-being. This will be assessed in terms of light, privacy, visual intrusion, internal space standards, noise, smell, pollution, and light pollution or other disturbances."*

Quality of the Public Realm

4.21 *"…A very high quality approach to the design of new, and works to existing, public realm and open space is required."*

4.22 *"Public Realms includes streets, hard and soft spaces and routes used by the public. The design of public space should take careful consideration of the proximity and accessibility to local residents and all public realm should be designed with the principles as set out under layout, form and functioning of development above."*

3.16 The second policy found in the High Quality Design and Sustainable Development is further emphasised within the South Marston and Eastern Village Supplementary Planning Documents, this policy is found below:

Policy DE2: Sustainable Construction

a. *"Development shall demonstrate passive solar benefits, (heating, ventilation, cooling and lighting) in accordance with Policy DE1, through the layout and design of the site, and orientation and design of buildings.*

b. *All major development... shall meet the following sustainable construction standards.*

Development type	***By April 2013***	***2013-2016***	***2016***
Residential	***Code for Sustainable Homes level 3 (In full)***	***Code for Sustainable Homes level 4 (In full)***	***Code for Sustainable Homes level 5 (in full)***
Non-Residential	***BREEAM Excellent***		

c. *Where on-site provision of renewable technologies is not appropriate, new development can meet CO2 reduction requirements through off-site provision by making "allowable solutions contributions". Funds gathered will be used for wider energy efficiency and energy generation initiatives.*

d. *The use of combined heat and power (CHP), and/or combined, cooling, heat and power (CCHP) with district heating is encouraged. Within the identified "District Heating Priority Areas" and strategic allocations, as shown on the Proposals Map, major development shall:*

- *Demonstrate a thermal master-planning approach considering issues such as mix of uses, anchor loads, density and heat load profiles to maximise opportunities for the use of district heating; and*
- *Incorporate infrastructure for district heating and connect to existing systems where and when they are available or fully justify any alternative approach."*

Figure 3b: Policy DE2 – Sustainable Construction

3.17 With regards to part (d.) in Policy DE2, Bizcat Consulting completed the 'District Energy Pre-Feasibility Study' on behalf of Swindon Borough Council. This study focuses on district energy whilst heading towards suitable supply systems due to the findings of heating and cooling demands. The report considered Swindon to be capable of providing promising opportunities for district heating infrastructure. The District Heating Priority Area is focussed on the town centre and principal employment areas and does not include the strategic allocations (including the New Eastern Villages).

3.18 In support of the local development framework Swindon Borough Council published the 'Infrastructure Delivery Plan' in December 2012 which was updated July 2013. This explains the formation of the informal energy partnership and briefly describes the alignment of strategic projects to integrate with a district heating system as mentioned previously.

3.19 The Swindon Central Area Action Plan outlines further policies on sustainability albeit in the town centre only with 'Policy CAAP2: Sustainable Construction' highlighting the requirement for sustainable waste management, this have been outlined below:

"Major development proposals must demonstrate that steps will be taken to:

- *Reduce, re-use and recycle any waste produced through the development process;*
- *Minimise the use of raw materials in the development process;*
- *Minimise the pollution potential of any unavoidable waste;*
- *Dispose of any unavoidable waste in an environmentally acceptable manner; and*
- *Ensure maximum waste recovery once the development is completed."*

3.20 This report will therefore explore the options currently available to the development in achieving national and local policy requirements mentioned within this section as well as the potential wider sustainability benefits.

Fuel Poverty

3.21 It is currently estimated that over four million homes in the UK are considered to be in fuel poverty. Fuel Poverty Households are considered to be in fuel poverty when more than 10% of the household income is spent on fuel to keep the home in a "satisfactory condition".

3.22 Fuel Poverty is caused by the influence of three main factors:

- Low income
- High fuel prices
- Poor energy efficiency

3.23 Recognising this as an important issue, this document outlines potential low carbon/ renewable technology and/or fabric improvements that significantly increase energy efficiency and reduce energy requirements as per the requirements of current policy to ensure all future residents are able to achieve reduced fuel bills.

4 Development Sustainability/Energy Proposals

4.1 As outlined above a review of relevant local and national policy sustainability requirements has been completed to ensure that the new development offers feasible/suitable measures to accommodate and enhance its sustainability credentials.

4.2 **Base Proposals:** Compliance with Code for Sustainable Homes level 3 / Building Regulations.
Possible Option: Incorporation, integration or delivery of district heating and/or combined heat and power where viable.

4.3 The sustainability measures will deliver a development that has a low environmental impact in keeping with local requirements and national standards. Chapter 6 'Sustainability Benefits' covers in more detail the associated benefits not only to the future residents of the Phase I Swindon New Eastern Village development, but to the wider areas covering benefits to existing residents.

4.4 Sustainability has been at the core of all actions from early masterplanning to detailed design, construction, and management of the completed development. Sustainability applies to the way people live as well as the way places are planned and built. The site has been designed to give people the opportunity to choose a healthier, more sustainable lifestyle; and to allow people to lead their lives differently: reducing energy, water consumption and waste production, and creating an inclusive place that engenders social connections.

5 Code for Sustainable Homes

Code for Sustainable Homes

5.1 To fully understand the implications of Code for Sustainable Homes it is important to outline the necessary requirements. The Code assesses nine categories in relation to sustainable design:

- Energy and CO_2 Emissions
- Water
- Materials
- Surface Water Run-off

- Waste
- Pollution
- Heath and Well-being
- Management
- Ecology

5.2 The development will achieve Code for Sustainable Homes Level 3 in all properties. To achieve Level 3 a property must score 57%, or more, from the total points available within the range of categories identified above. The following information briefly outlines the likely improvements and mandatory standards that will be required to achieve Level 3.

Energy and CO_2 Emissions

5.3 This document goes on to specify how the development at Phase I Swindon EV will achieve compliance with the mandatory requirements for Code Level 3 and with Building Regulations Part L: 2010 in all dwellings, through a selection of sustainable construction techniques, to achieve the respective reductions in CO_2. Figure 5a outlines the mandatory minimum standards in the reduction of CO_2 emissions.

Table 1.2: Code Levels for Mandatory Minimum Standards in CO_2 Emissions (Ene 1)

Code Level	Minimum Percentage Improvement in Dwelling Emission Rate over Target Emission Rate
Level 1 (★)	0% (Compliance with Part L 2010 only is required)
Level 2 (★★)	0% (Compliance with Part L 2010 only is required)
Level 3 (★★★)	0% (Compliance with Part L 2010 only is required)
Level 4 (★★★★)	25%
Level 5 (★★★★★)	100%
Level 6 (★★★★★★)	Net Zero CO_2 Emissions

Figure 5a: CO_2 Reduction Requirements for CfSH (http://www.communities.gov.uk)

5.4 These reductions in CO_2 can be achieved through energy efficiency measures that may include fabric improvements as well as low carbon / renewable technologies such as photovoltaics (solar panels), air/ground source heat pumps. The renewable energy options are discussed later in this report.

5.5 It is feasible to reduce CO_2 generation through improvements in building fabric, which in turn achieves better thermal performance. To achieve the CfSH Level 3 requirements, it is possible to implement fabric improvements alone, across much if not all of the proposed development.

5.6 The following building fabric enhancements are viable measures to provide the thermal properties needed to help achieve Code for Sustainable Homes requirements:

- Mechanical ventilation with heat recovery
- Cavity wall improvements
- Improved insulation in the roof and flooring
- Better quality windows and doors
- Minimisation of thermal bridging
- Improvement of air permeability
- Minimum 75% LED / low energy lighting
- Advising and promoting the use of more energy efficient products
- Gas condensing boiler with interlock and weather or load compensator
- 10% Secondary heating system

5.7 The development will achieve the required level of CO_2 reductions as part of meeting the Code for Sustainable Homes Level 3 criteria across all dwellings.

Water

5.8 To achieve Code for Sustainable Homes Level 3, properties will have measures introduced to restrict water consumption to a maximum of 105 litres per day per person. This requirement will be achieved through the use of water efficient fittings, such as delayed inlet valves and flow restrictors, and water efficient appliances, such as low flush WC's.

Materials

5.9 Compliance with Level 3 requires three or more of the following building envelope materials to achieve a rating of A+ to D, as per The Green Guide.

- Roof
- External walls
- Internal walls
- Upper and ground floors
- Windows

5.10 The Green Guide assesses the materials upon their environmental impact and performance, which leads to improved thermal properties allowing a reduction in energy requirements.

5.11 Materials for the construction of the dwellings will be responsibly sourced and will minimise environmental impact.

Surface Water Run-off

5.12 The development will employ the use of best practice on-site drainage systems to ensure that the site provides no detrimental increase in surface water run-off, to ensure that any significant environmental impact is mitigated.

Code for Sustainable Homes Non Mandatory Credits

5.13 By meeting the mandatory standards for Level 3 the development will achieve a minimum of 4.7%, of the required 57%, indicating that a further 52.3% will be required to be awarded Code for Sustainable Homes Level 3.

5.14 These additional credits will be achieved through meeting further targets within the aforementioned nine categories. The development has been designed to ensure that provision has been made to include these further elements, which may include:

- Further energy efficiency
- Energy labelled white goods
- Drying space
- External lighting
- Home offices, including broadband provision
- Storage for recyclable and non-recyclable waste
- Waste management plans
- Home user guides
- Low or zero carbon energy sources

5.15 As a result of implementing Code for Sustainable Homes Level 3 across the site, the proposed development will meet local policy requirements.

6 Sustainability Benefits

6.1 The site offers a number of sustainability measures through the implementation of sustainable development proposals. This chapter of the report reviews the wider sustainability benefits that the development will provide to the future residents and service users as well as the added value the site will bring to the local community.

Design

6.2 A high quality of design throughout the development, including new streets, open spaces, together with ongoing maintenance will help to promote respect towards the environment and therefore increase its use and safety and the overall sustainability benefits of the site.

Energy

6.3 This report outlines measures in which the development will look to incorporate low carbon / renewable energy measures and increased efficiency in line with local and national policy.

6.4 The developer's desire to deliver a high quality settlement with enhanced living standards tackling the issues of CO_2 emissions and fuel poverty will be underpinned comprehensively by Code for Sustainable Homes level 3 integrated into each property.

6.5 The proposals provide leading sustainability credentials across the development that will result in tangible benefits to the environment and to occupiers of the development. The proposals substantially reduce the demand on fossil fuels and associated CO_2 production, all corresponding to significant reductions in household bills.

Water Resources

6.6 As mentioned previously, measures will be introduced to restrict water consumption to a maximum of 105 litres per day per person.

6.7 Considering the onsite sustainability proposals, it is anticipated that a typical property at Phase I Swindon will require approximately 114,975 litres of water a year.

6.8 OFWAT confirm that for the period of 2009-2010 the average water consumption in the UK equated to 143.2 litres per person per day.

6.9 The following table, Figure 8a, compares the water requirements for a typical UK housing stock home against a home at the proposed development.

	Typical Annual Water Consumption (litres)
Typical UK home	156,804 litres[1]
New home at Phase I Swindon	114,975 litres

Figure 6a: Typical annual water usage for properties at Phase I Swindon

6.10 The table outlines that a typical property at the development will offer a minimum of a 26% reduction in the water usage from that of a typical existing UK home.

[1] OFWAT confirm that for the period of 2009-2010 the average water consumption in the UK equated to 143.2 litres per person per day.

6.11 This strategy will provide wider benefits to the local areas by ensuring water demand is substantially reduced and that water resources are capable of accommodating the demands of the site and surrounding area.

Sustainable Drainage

6.12 The use of Sustainable Urban Drainage will help reduce the volume and rate of surface water run-off from the site. The drainage strategy will employ storage methods such as ponds and in order to attenuate flow, remove pollutants, and ensure that runoff to existing watercourses is restricted according to the existing hydrology of the site. All SuDS elements will be sized to accommodate increased flows from potential future climate change.

6.13 Through the use of permeable hard surfaces and attenuation ponds, the site-wide water management and drainage solution will avoid any increase in surface water run-off to the natural watercourses, with the proposals set out in the Flood Risk Assessment significantly reducing the level of surface water runoff from the development.

Waste

6.14 The approach to delivering a sustainable waste management system will begin inside buildings by providing ample internal segregated waste storage space inside residential properties. External storage for recycled waste will be well considered, and integrated into the building envelope or boundary condition to ensure an attractive solution to home-collection of waste for recycling.

6.15 Consultation with the local authority and the local waste management contractor will determine the detailed approach to delivering waste minimisation and high levels of recycling. During construction full consideration will be given to minimising waste, promoting reuse and recycling materials.

Transport

6.16 The sustainability objectives for the provision of transportation and transport facilities focus on the following priorities:

- Reduce the carbon emissions associated with transport by reducing car dependency for local journeys
- Provide sufficient convenient and safe facilities for walking and cycling within the site and into the town centre
- Facilities to encourage cycling, such as road designs incorporating safe cycle paths, and buildings including cycle parking.

6.17 The site occupies an excellent location for development to be sustainable, being close to established residential areas, employment and a range of facilities including a primary school and local shops.

6.18 Sustainable transport such as walking and cycling have been given priority in the development, with the structure providing legible and direct routes that follow desire lines.

6.19 Links to existing public rights of way have been incorporated into the site and new footpaths and cycle routes are proposed connecting to those, reinforcing the sustainable transport and pedestrian links. Movement for pedestrians and cyclists is fully integrated into the site with designated paths.

Well Being and Healthy Living

6.20 The development layout and facilities have been considered to ensure that the site offers:

- A compact and well designed layout providing recreation and exercise space for all within a five minute walk
- Promote walking and cycling to destinations using new and existing foot and cycle links

- Provision of multifunctional green spaces through a well connected green network easily accessible to all
- Enhance Bio-diversity by retaining existing trees vegetation where possible on site and integrate with new green infrastructure network
- Provision of spaces that encourage sustainable living and sense of community pride and ownership.

7 Energy Statement

7.1 Swindon Borough Council, within Policy DE2: Sustainable Construction, encourage combined heat and power (CHP), and/or combined, cooling, heat and power (CCHP) to be included as part of the encouraged district heating strategy outlined within the 'Pre-Feasibility' study. However, it is worth noting that the provision of district heating can result in a significant increase in development implementation costs not least in advance of the implementation of a town wide scheme. These costs are reduced where a long concession Energy Services Company (ESCO) is linked to the delivery, however, this is not established at present. Therefore at this stage, the suitability of a district heating/ combined heat and power system has to be subject to careful viability testing.

7.2 This section of the report reviews the likely energy use and CO_2 production for the development based upon the basic Building Regulation standards and how this is affected by the proposed implementation of Code for Sustainable Homes and Renewable Energy.

7.3 The CO_2 emissions have been calculated based on the following conversion factors:

	CO_2 Conversion Factors (kg/MWh)
Electricity	517
Gas	198

Figure 7a: CO_2 Conversion Factors (SAP 2009)

Residential Development

7.4 Energy requirements for the residential development have been calculated using a total energy requirement factor of 0.12MWh/m^2. This figure is based upon industry accepted information on energy demands.

7.5 For the purposes of this report the breakdown of residential units has been assumed. Figure 6b, summarises the estimated energy use and CO_2 generated if the proposed development were to proceed without any improvement over UK Building Regulations Part L. The breakdown demonstrated within the figures below is for the purpose of example and the actual mix is likely to vary.

Bedrooms	Total	Assumed GIFA (m^2)	Total Area (m^2)	Space Heating (MWh)	Water (MWh)	Electricity (MWh)	Cooking (MWh)	Total Energy (MWh)	CO_2 Annual Emission (kg)
1	71	65	4,615	111	222	138	83	554	153,818
2	881	70	61,670	1480	2960	1850	1110	7,400	2,055,461
3	762	80	60,960	1829	2560	1829	1097	7,315	2,031,797
4	595	100	59,500	2142	2142	1785	1071	7,140	1,983,135
5	71	110	7,810	281	281	234	141	937	260,307
Total	**2,380**		**194,555**	**5,843**	**8,165**	**5,836**	**3,502**	**23,346**	**6,484,518**

Figure 7b: Residential Baseline Energy Uses

7.6 It can be seen a total annual energy demand of 23,346 MWh and a CO_2 generation of 6,484,518kg (6,484 tonnes) is predicted across the site without any improvement over Part L of the Building Regulations. An average energy demand of 9,809 kWh/yr per unit with average CO_2 emissions of 2,725kg per unit.

Low Carbon / Renewable Energy

7.7 Whilst Swindon Borough Council emphasise the requirement for district heating/combined heat and power, a large range of alternative low carbon/renewable energy options are available for new development. However, when specifying a system, it is necessary to have regard to a range of considerations, such as:

- Energy demand and supply
- Cost of implementation and payback
- Visual and space / land use characteristics
- Supply of any raw materials
- Maintenance and reliability

7.8 To support growth in renewable electricity generation, Feed-in-tariffs (FITs) have been implemented within the UK energy market, making renewable energy systems that generate electricity a more appealing option to developers and home owners. The FITs have been in effect since April 2010.

7.9 FIT allows properties with renewable energy systems to generate an income from the energy produced as well as selling surplus energy back to the network provider by feeding it back into the Grid. The introduction of the FITs has signalled the end of all new applications for electrical microgeneration grants from the government. FITs are tax free and will be paid over a minimum of 10 years from the date the system is registered.

7.10 The adoption of Feed-in-Tariffs has improved the financial viability of certain technologies, particularly photovoltaic's, which were previously a costly solution in achieving Renewable Energy or CO_2 reduction criteria.

7.11 Additionally a further generation tariff is being introduced in relation to renewable technology that meets heating requirements. It is anticipated that the Renewable Heat Incentive will be introduced on a domestic level as of late 2013, this will work on the same basis as the FIT's by which the properties fitted with the technology will receive an income from each kWh generated.

7.12 The relatively recent introduction of FiT's has brought about a step change in the use of Solar Voltaic panels on many existing properties and new developments, this now being considered commonly as the preferred technology.

7.13 Other potential options are:

- Solar Water Heating
- Micro CHP
- Ground and Air Source Heat Pumps
- Biomass
- District Heating Systems

7.14 The potential beneficial technologies are reviewed below.

7.15 **Photovoltaic (PV):** PV modules are available in a large variety of forms, including roofing tiles and glazing panels. These systems convert daylight into electrical currents that may be used to provide power to a wide range of applications.

7.16 PV systems look similar to flat plate collectors, although they provide a better energy return. A typical PV system is shown on Figure 6c.

7.17 PV systems, particularly roof tiles, present an innovative and deliverable approach to delivering renewable energy. Advantages include the ability to be applied more flexibly and to create a better design solution than other technologies.

7.18 Also following on from the introduction of Feed-in-Tariffs, the application of PV has become a more attractive prospect to home owners as payback times have been reduced making the technology more feasible. However, PV still has a significant implementation cost which limits the potential for its use.

Figure 7c: PV system

7.19 **Solar Thermal Technology (Solar Water Heating (SWH)):** is currently one of the most cost efficient means of providing renewable energy to residential developments. One of the most widely recognised forms of SWH is the "Flat Plate Collector", (FPC). These are broad, exposed solar irradiation absorbing panels that are commonly fitted on the roofs of residential properties, as shown on Figure 6d and 6e below. The systems collect solar energy and convert this directly into heat, which is generally used for the supply of hot water in residential properties. Alternatively, "Evacuated Tube Collectors" (ETC) can be employed; being generally thin glass tubes coupled with a heat exchange system that improves the thermal performance of the system.

7.20 FPC and ETC's technologies provide a more cost effective and versatile addition to residential energy supply than many of the other renewable energy techniques. Although ETC systems are more costly than FPC's, they offer a more efficient and reliable source of renewable energy as they are less susceptible to poor supply of direct sunlight. Both of these options offer good capital return rate, with the system being likely to pay back the cost of installation in a minimum of approximately five years. The one minor drawback is that both of these systems require specific south east or south west orientation to maximise their efficiency.

7.21 The average peak output for a solar water heating panel on a typical housing installation is 900kWh/m^2.

7.22 For maximum efficiency, solar panels should be mounted on a south facing roof at a 30^0 angle with the horizontal and away from trees, surrounding buildings and chimneys. Fortunately, the average tilt of a UK house roof is about the optimum for receiving solar energy in the UK.

7.23 A typical installation would aim to provide 60-70% of the homes annual water heating requirement. An example of these system are shown in Figures 6d and 6e.

Figure 7d & 7e: Flat Plate Collector system and Evacuated Tube Collector system

7.24 **Micro Combined Heat and Power (Micro CHP):** is designed as a replacement for gas powered boilers and a supplement to mains electricity. Micro CHP units generate electricity by recycling the waste heat from a conventional boiler heating process.

7.25 These systems rely on each boiler being fitted with the Micro CHP unit. While this is not a substantial drawback in terms of its operational value, it is generally not cost effective to implement over a large residential development. Micro CHP units can result in increased noise output and may require careful management in residential dwellings.

7.26 Baxi have now released a micro CHP unit called the 'Ecogen'. The unit, as seen in Figure 6f, costs significantly more than a traditional boiler heating system. Extensive field trials have been completed in over 400 UK homes and initial results have shown that Baxi Ecogen units can reliably satisfy up to two thirds of a typical household's electrical requirements.

7.27 This form of renewable technology will benefit from the new Feed-in-Tariffs and a typical Baxi Ecogen installation will be eligible to make 10p for every kWh generated and 3p/kWH for any energy exported.

Figure 7f: Baxi Ecogen

7.28 **Air Source Heat Pumps:** provide relatively 'low-grade' heat from the air, which is then transferred, via a compression tank, to the required source. The systems perform exceptionally well when paired with under floor heating systems.

7.29 Heat pumps require a feed of electrical energy to supply the pump which produces the heat energy. Most heat pumps produce an output based on a coefficient of performance (CoP) ratio of circa 3:1. For every 3kWh of heat energy produced 1kWh of electricity is used. Therefore, heat pumps can provide a solution to meeting Building Regulations 2010.

7.30 A typical 6kW system would aim to provide the entire heating requirements for most homes.

7.31 **Ground Source Heat Pumps:** operate in a very similar way to ASHP; however the heat energy is sourced from the ground and has an improved CoP of up to a ratio of 4:1 when compared with air source heat pumps, providing a more efficient source of heat energy than that of ASHP. The initial cost outlay tends to be significantly higher as the systems need to be installed into the ground via boreholes or alternative below ground installations. On most sites this constraint proves prohibitively costly.

7.32 **Biomass Boilers:** These systems operate using an array of bio fuels such as recycled wood or virgin chippings. They can be implemented within individual properties but are often used in larger systems as a renewable energy hub for a parcel of development.

7.33 While it is technically feasible to implement biomass in residential development, the physical size of the equipment and raw material supply demands best suit installation in communally heated units or commercial developments, therefore it is considered appropriate to implement such technology within the commercial areas.

7.34 Initial installation costs are significantly greater to that of conventional gas heating systems.

7.35 However there is significant grant funding now available to offset the initial cost increase associated with Biomass systems making Biomass a feasible and affordable technology to be implemented where possible.

7.36 **District Heating Systems:** These systems rely on a centralised district heating centre in which a biomass or other fuel system with a back up gas supply provides the heating to properties. It is then distributed by a network of pipes, together with electricity generated from the heat output.

7.37 District Heating Systems are best suited to providing the base central heating load for the development (hot water supply), with conventional gas boilers responding to the peak demands (winter space heating). These systems are not always capable of effectively meeting the demands of peak loading periods, however, the Bizcat 'District Energy Pre-Feasibility Study' reports that very promising opportunities are present for introducing and expanding district heating systems within the Swindon area.

7.38 However, at present, there is no district heating system present in Swindon with progress towards its implementation in central areas and in principal employment areas uncertain. Therefore, it is not demonstrated that district heating is a viable option in the context of the application.

Energy Options

7.39 This document sets out a range of sustainability measures that can be provided at the proposed development that meets national and local policy requirements. It is important to provide an array of viable options/solutions for the development to ensure that a suitable strategy is available in an ever changing/evolving market.

7.40 The benefit to providing a flexible sustainability strategy is to ensure that the developer is able to respond positively to the issues of energy generation and CO_2 reduction. Further advances in technology may determine that in the future alternative solutions are more suitable than that of the current recommendations.

7.41 Measures covered in new and upcoming policy such as 'Allowable Solutions' could, in the future, provide cost effective and viable ways of achieving a similar or improved strategy.

8 Summary

8.1 This document sets a range of sustainability measures that can be implemented at the proposed development to meet national and local policy requirements in accordance with the guidelines in NPPF whilst maximising and adding additional value to the local growth areas.

8.2 The proposals provide good sustainability credentials across the development that will result in tangible benefits to the environment and to occupiers of the development. Along with reductions in water usage, the proposals substantially reduce the demand on fossil fuels and associated CO_2 production, all corresponding to significant reductions in household bills.

8.3 Sustainability will be at the core of the development proposals ensuring that the site is designed to give people the opportunity to choose a healthier, more sustainable lifestyle. The site has been designed to allow people to lead their lives differently: reducing energy, water consumption and waste production, whilst providing additional transport links and facilities to the wider local area.

8.4 The sustainability market is rapidly evolving. Technology is advancing at a pace that may result in currently favourable options being less appropriate in a relatively short period. At the time of implementation, alternative means of installation may be preferable to achieve the required amount of renewable energy generation / carbon reductions and sustainability measures.

9 Limitations

9.1 The conclusions and recommendations contained herein are limited to those given the general availability of background information and the planned usage of the site.

9.2 Third party information has been used in the preparation of this report, which Brookbanks Consulting Ltd, by necessity assumes is correct at the time of writing. While all reasonable checks have been made on data sources and the accuracy of data, Brookbanks Consulting Ltd accepts no liability for same.

9.3 The benefits of this report are provided solely to Hallam Land Management Ltd for the proposed development land at Phase I Swindon New Eastern Village.

9.4 Brookbanks Consulting Ltd excludes third party rights for the information contained in the report.

This Page Is Intentionally Blank